

Integrating Dynamic Difficulty Adjustment and Alien Motion in Virtual Reality for Cognitive Rehabilitation Therapy

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Abstract

Virtual reality has been shown to have a positive impact on the effectiveness of cognitive rehabilitation, as well as improving user engagement and motivation during the process. Operators have full control over the virtual environment, allowing them to adjust the task difficulty or manipulate the simulation to assist the user during the therapy. Adjusting the difficulty and supporting users through alterations could enhance virtual training effectiveness. Our research aims to investigate how the combination of these two methods can enhance rehabilitation outcomes, both in terms of objective results and the psychological state of the users.

CCS Concepts

• **Applied computing** → *Interactive learning environments*; • **Human-centered computing** → **Virtual reality**; • **Computing methodologies** → **Virtual reality**.

Keywords

Virtual Reality, Sense of Embodiment, Dynamic Difficulty Adjustment, Alien Motion, Self Efficacy

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1 Introduction

Cognitive function disorders are a common consequence of various types of neurological damage. They can affect language, memory, perception, and executive functions, leading to a significant reduction in quality of life [23]. Patients with cognitive impairment often have to go through a cognitive rehabilitation process. This involves the application of various procedures and support techniques to help them live safely, productively, and independently in their environment [24]. One of the problems with this process is that it requires large amounts of exercise to be effective. It must be intensive and prolonged in time, and the patient must be motivated and

engaged throughout the whole duration of the treatment [6, 23]. Among the techniques used for cognitive rehabilitation, Virtual Reality (VR) seems to be very promising as it allows the patient to interact with a virtual environment specifically designed to support the recovery of cognitive functions [22]. This technology is defined as a medium that consists of interactive computer simulations. Users are provided with feedback related to one or more senses (in most cases vision and hearing) based on their position and actions within the virtual environment. This way, users develop the feeling of being immersed in the experience [30]. A key element in the effectiveness of VR-based training is the system's capability to adapt to the patient's ability level and needs [38]. An effective adaptive system should incorporate concepts of adaptive variable, performance measure, and adaptive logic [19]. The adaptive variable is a dynamic attribute adjusted according to the performance measure following an adaptive logic. Several works suggested that the tasks' difficulty can be a significant adaptive variable for improving the efficacy of training and users' motivation [4, 15, 16, 18]. Dynamic Difficulty Adjustment (DDA) is a technique that consists of adapting the tasks' difficulty based on the user's capabilities in real-time [39]. DDA methods can be categorized into two distinct approaches: rule-based and data-driven. 1) Rule-based techniques rely on predefined rules established by the system designer, typically incorporating performance metrics or additional parameters to regulate the difficulty level. In contrast, 2) data-driven systems employ statistical or machine learning methodologies to adjust the level of difficulty. Data-driven methods include Reinforcement Learning (RL), a technique that can make decisions in uncertain situations by taking actions and trying to maximize a reward. In the DDA context, the RL agent could adjust the difficulty as an action, trying to maximize the users' performance [25]. Using RL as adaptive logic could be a valid option in the domains where it is difficult to have a large dataset of training data, such as rehabilitation [38]. A further potential adaptive variable could be the feedback provided to inform users of their real-time performance and movement within the experience. Feedback can be adjusted to encourage the patient and facilitate the training [38]. In this scenario, deliberately providing trainees with false feedback about their performance levels (without their awareness) can improve motivation and training outcomes [1, 27]. We will refer to these alterations with the term alien motion, indicating a modification of the virtual body or in the behavior of the virtual environment without the users being aware of it [31–34]. To the best of our knowledge, using a combination of these two adaptive variables (difficulty and alien motion) is still unexplored and can be a promising approach to enhance a VR-based training process. The combination of objective and subjective

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factors is the most common method of evaluating a training system, as it can capture more features that could be useful in improving training effectiveness [35]. Subjective factors include self-efficacy, defined as the belief people have about their capabilities in producing certain performance levels. This cognitive process determines how much effort a person will employ in a certain task, how much it will be sustained, and the ability to cope with difficulties [2, 3]. Self-efficacy is not widely used as an evaluation criterion for virtual training systems, although it is a key element in enhancing traditional cognitive rehabilitation outcomes [7, 11, 36]. Performance accomplishment is among the factors that increase self-efficacy. Thus, users should experience success to increase their belief in their abilities. On the other hand, repeated failures may negatively influence that confidence [2, 3]. We believe that dynamically adjusting the difficulty and simultaneously supporting users through alien motion during rehabilitation without their awareness may increase their self-efficacy and improve rehabilitation outcomes. Another subjective element that we are considering for the system evaluation is the Sense of Embodiment (SoE). This perception is a key factor in virtual rehabilitation [37] and it refers to the feeling of having and controlling a virtual body [20]. In the design of the VR rehabilitation system, it is crucial that developers and scientists provide an experience that does not negatively affect the user's SoE. We believe that a combination of 1) objective performance, 2) self-efficacy, and 3) SoE could represent a valid measure for evaluating a VR-based training system.

2 Related Works

VR technology has been widely used and explored in the context of cognitive rehabilitation. Several works investigated its effectiveness with promising results on post-stroke [13, 14], multiple sclerosis [28], or Parkinson patients [5]. To increase the effectiveness, a VR training system should adapt to the patient's needs (and thus be adaptive) and one of the problems in this scenario is to find the adaptive variable [38]. While the majority of the adaptive VR systems use the difficulty of the tasks as an adaptive variable [8, 12], another interesting possibility is to introduce alteration within the virtual experience, to provide users with false feedback and support them during rehabilitation without them being aware of it. Soccini *et al.* [31–34] investigated the relationship between the physical and the virtual body. In particular, the user's ability to adapt when a gap is introduced between real and virtual body movement (without the user being informed) was analyzed. The results of this research showed that despite the alterations, the users still felt control over their actions and movements. Inamura *et al.* [17] introduced alterations in terms of the size and appearance of the arm during a virtual rehabilitation task. This study showed no negative influence on the users' perception of controlling their movements. These studies demonstrated that it is possible to introduce alien motion within a VR experience without compromising the users' perception that they are the ones performing the actions. Among the key factors for positive traditional rehabilitation outcomes, there is self-efficacy, and the impact of adaptive virtual training on this cognitive process is still a novel topic. Lui *et al.* [21] explored the effect of adaptive training on outcomes and self-efficacy in a fine motor task. There was no significant difference in self-efficacy between two different

levels of adaptiveness, indicating the need for further exploration. Nevertheless, there was an improvement in performance in both conditions, indicating VR as a promising tool in the training context. Regarding the adaptive logic, reinforcement learning seems to be the best option in the rehabilitation context given the lack of data at the start of the training [38]. Rahimi *et al.* [29] proposed a reinforcement learning-based DDA approach in a visual working memory game. The RL model's action consists of setting the appropriate difficulty level, while the state is represented by a vector containing the difficulty and the user's score. This technique was compared with two rule-based DDA methods and was found to be the most effective in enhancing the game experience and achieving higher user scores.

3 Goals and Research Questions

The final goal of our research is to define and validate a new approach for enhancing a cognitive rehabilitation process by using VR technology. The proposed method consists of a combination of two elements: 1) The introduction of **alien motion** to provide users with false feedback and to support them when necessary, avoiding exposure to repeated failures; 2) Adapting the **difficulty** of the training tasks according to the users' capabilities and needs. To evaluate the effectiveness of this approach, we will consider not only the objective outcomes of the training but also subjective elements, namely self-efficacy and sense of embodiment. Consequently, these three factors will be used as the measures to consider to adjust the extent of difficulty and alien motion. Regarding these goals, the research questions that are guiding this work are:

- **RQ1.** To what extent does supporting people through alien motion during a training task influence SoE, self-efficacy, and outcomes?
- **RQ2.** What are the relevant training measures to use for dynamically adjusting difficulty and alien motion?
- **RQ3.** What is the best adaptive logic to use for dynamically adjusting difficulty and alien motion?

4 Innovative contribution

Several studies have explored the effects of alien motion by introducing spatial or temporal gaps between the movements of the virtual body and the user's real body. However, our approach involves modifying the behavior of the objects that the user interacts with. To the best of our knowledge, alien motion and DDA techniques are usually applied separately. The proposed approach investigates the integration of these two techniques. Furthermore, although the relationship between self-efficacy and rehabilitation outcomes in the real world has been widely studied, the same relationship in the virtual world is still a novel topic. Therefore, we also aim to investigate this aspect. Such methodology could represent a step forward in the design of user-centered VR experiences. Future human-computer interaction designers may consider supporting users in order to provide implicit guidance towards a better and more satisfying experience, not only in rehabilitative contexts but also in gaming, sport, and education. In addition, exploring the use of machine learning techniques in these cases of human-computer interaction could help make these systems more accessible by making them adaptable to a wider range of users.

5 Preliminary studies and results

We developed a VR training system by considering a simple task: hit the centre of a target by throwing a ball. Figure 1 shows a user executing the launch movement. The system can support users without them being aware of it by altering the ball's trajectory (Figure 2). We tested three different support management strategies:

- **No support:** The ball trajectory is not altered;
- **High static support:** The user receives a high level of support that does not change over time;
- **Dynamic support:** The level of support varies according to the accuracy of recent throws, assisting only when the user's shooting quality is poor.

So far, we conducted two different experiments on 30 and 18 participants respectively to assess the relationship between these support management strategies and 1) Sense of Embodiment, 2) Objective performances, 3) Enjoyment, and 4) fatigue, both mental and physical.

5.1 Influence on Sense of Embodiment

By using this training system, we initially investigated the influence of supporting users through alien motion on SoE. Generally, the introduction of support did not have a negative effect; on the contrary, it led to small improvements. Users who received high static support experienced a decline when the support was removed. Dynamic management resulted as the most effective method as it led to improvement upon implementation and limited the decrease when support was removed. In conclusion, our results showed that dynamic support has minimal impact on the participants' perception of controlling their virtual bodies and actions. Despite the introduction of these alterations, the SoE mean value was always positive in all sessions [9].

5.2 Influence on Objective Outcomes

Secondly, we explored the support's impact on the users' objective performances with promising results. Performance was quantified as the percentage of successful shots and its variation from the beginning to the end of the training. We found that unsupported users were the only ones to experience a drop in performance from the first to the last session. The findings of the present study showed the efficacy of dynamic support in comparison to the other two methods. In particular, users subjected to adaptive training exhibited a positive change in performance, with an increase of almost 10% from their initial to final performances [10].

5.3 Influence on Enjoyment and Fatigue

Finally, we measured and compared enjoyment, and mental and physical fatigue between groups. Almost the totality of the participants evaluated the virtual training experience as enjoyable, independently of the group. No significant differences in physical or mental fatigue were observed among the groups, with mental fatigue being particularly low. [26].



Figure 1: User performing a launch during a system testing session

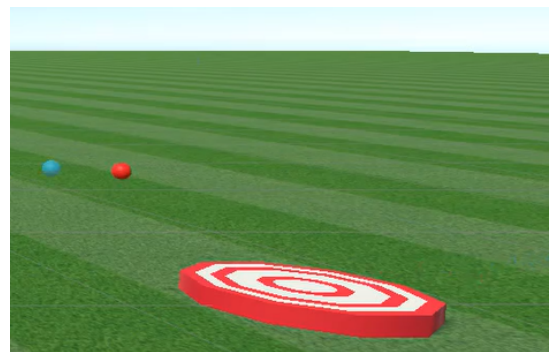


Figure 2: Difference between the real trajectory(blue ball) and the enhanced trajectory(red ball)

6 Future work

The results obtained to date are only the first step in this project. In future research, we will attempt to answer the previously mentioned research questions.

- First, we will use the BallThrow VR training system to test the influence of support on users' self-efficacy;
- Second, we will implement a data-driven model that manages support according to the needs of individual users, providing greater adaptability and flexibility than the current rule-based method. We believe that RL could represent the best option for adjusting the amount of alterations in real-time as it allows making decisions in uncertain situations, without the need for a large amount of initial training data;
- Third, a DDA algorithm will be integrated into the system to adjust the difficulty of the tasks according to the user's ability. Difficulty management will also be implemented using a RL model;
- Fourth, we will apply this approach given by the combination of alien motion and DDA to cognitive rehabilitation tasks such as activities of daily living, attention, memory, or upper limb exercises.

We also plan to apply this methodology also to other contexts where the goal is to develop or train a particular behavior, skill, or knowledge, such as health, sport, and education.

7 Discussion and Conclusions

Using virtual reality in cognitive rehabilitation can bring improvements in both outcomes and patients' motivation. This technology makes it possible to dynamically adapt the tasks' difficulty to the users' ability and to support them by altering the simulation. This project aims to define and validate a methodology for virtual rehabilitation systems that consists of a combination of these two techniques. The evaluation of this approach will also consider the psychological sphere of the users, investigating the effects on their self-efficacy. So far, we developed a system that can modify the performance of users by altering the visual feedback they perceive, without their knowledge. In our experimental settings, participants were supported by this system during a target shooting training task. We investigated the effect of different support strategies on several factors. We found that dynamically supporting users could be the best option as it does not negatively affect the sense of embodiment upon its removal. Furthermore, people subjected to dynamic support were those who showed the greatest improvement in the number of successful hits from the start to the end of the training. Regardless of the group, people found the experience enjoyable and there were no differences in mental or physical fatigue between the conditions. However, participants reported a positive and not mentally tiring experience, suggesting the potential of virtual reality technology in the context of training. The next steps will include the influence of alien motion on self-efficacy, and the implementation of a reinforcement learning model to manage both the support and the difficulty amount. These initial results are promising and can contribute to understanding how the characteristics of virtual reality can be used in training contexts. In particular, the effect of the interaction between the users and the virtual environment could be designed to support and guide them towards a specific goal without their awareness. The ability to use interactions based on gestures and body movements, together with the control that operators have over the experience, makes this technology particularly suitable for human-computer interaction. Exploring the potential of virtual reality could be crucial to the design of future interactive systems.

Doctoral programme details

Vittorio Fiscale is currently enrolled in the second year of the PhD program in Computer Science at the University of Torino, Italy. His supervisor is Professor Agata Marta Soccini and his co-supervisor is Professor Federica Cena. This PhD program lasts three years and the expected completion date is by the end of 2026. He never attended a Doctoral Consortium before. The current submission was presented in a Doctoral Consortium for the first time at ACM CHI 2025.

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References

- [1] Belén Rubio Ballester, Martina Maier, Rosa María San Segundo Mozo, Victoria Castañeda, Armin Duff, and Paul F. M. J. Verschure. 2016. Counteracting learned non-use in chronic stroke patients with reinforcement-induced movement therapy. *Journal of NeuroEngineering and Rehabilitation* 13, 1 (Aug. 2016). doi:10.1186/s12984-016-0178-x
- [2] Albert Bandura. 1977. Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review* 84, 2 (1977), 191–215. doi:10.1037/0033-295x.84.2.191
- [3] Albert Bandura and Sebastian Wessels. 1997. *Self-efficacy*. Cambridge University Press Cambridge.
- [4] James William Burke, Michael McNeill, Darryl Charles, Philip Morrow, Jacqui Crosbie, and Suzanne McDonough. 2009. Serious Games for Upper Limb Rehabilitation Following Stroke. In *2009 Conference in Games and Virtual Worlds for Serious Applications*. IEEE, 103–110. doi:10.1109/vs-games.2009.17
- [5] Pablo Campo-Prieto, José Mª Cancela-Carral, and Gustavo Rodríguez-Fuentes. 2023. Immersive Virtual Reality Reaction Time Test and Relationship with the Risk of Falling in Parkinson's Disease. *Sensors* 23, 9 (May 2023), 4529. doi:10.3390/s23094529
- [6] Jimmy Choi and Elizabeth W. Twamley. 2013. Cognitive Rehabilitation Therapies for Alzheimer's Disease: A Review of Methods to Improve Treatment Engagement and Self-Efficacy. *Neuropsychology Review* 23, 1 (Feb. 2013), 48–62. doi:10.1007/s11065-013-9227-4
- [7] Keith D. Cicerone and Joanne Azulay. 2007. Perceived Self-Efficacy and Life Satisfaction After Traumatic Brain Injury. *Journal of Head Trauma Rehabilitation* 22, 5 (Sept. 2007), 257–266. doi:10.1097/01.htr.0000290970.56130.81
- [8] Pedro Henrique Barcha Correia. 2024. Adaptive Virtual Reality Solutions: A Literature Review. In *Symposium on Virtual and Augmented Reality (SVR 2024)*. ACM, 1–10. doi:10.1145/3691573.3691574
- [9] Vittorio Fiscale, Tetsunari Inamura, and Agata Marta Soccini. 2023. Enhancing Training and Learning in Virtual Reality: The Influence of Alien Motion on Sense of Embodiment. In *International Conference on Human-Agent Interaction (HAI '23)*. ACM, 412–414. doi:10.1145/3623809.3623944
- [10] Vittorio Fiscale, Tetsunari Inamura, and Agata Marta Soccini. 2024. Adaptive Training in Virtual Reality Through Dynamic Alien Motion Support. In *2024 IEEE Gaming, Entertainment, and Media Conference (GEM)*. IEEE, 1–5. doi:10.1109/gem61861.2024.10585746
- [11] Michelle Genis, Paul M. Camic, and Martin Harvey. 2015. Anxiety related to discharge from inpatient neurorehabilitation: Exploring the role of self-efficacy and internal health control beliefs. *Neuropsychological Rehabilitation* 26, 2 (Feb. 2015), 191–215. doi:10.1080/09602011.2015.1009473
- [12] Yusuke Goutsu and Tetsunari Inamura. 2024. Effectiveness of Adaptive Difficulty Settings on Self-efficacy in VR Exercise. In *30th ACM Symposium on Virtual Reality Software and Technology (VRST '24)*. ACM, 1–2. doi:10.1145/3641825.3689690
- [13] Hendry Gunawan, Irene Gunawan, Yetty Hambarsari, Rivan Danuaji, Baarid L Hamidi, and Benedictus Benedictus. 2024. Virtual reality intervention for improving cognitive function in post-stroke patient: A systematic review and meta-analysis. *Brain Disorders* 15 (Sept. 2024), 100152. doi:10.1016/j.dscb.2024.100152
- [14] Martha Gustavsson, Emma K. Kjörk, Mattias Erhardsson, and Margit Alt Murphy. 2021. Virtual reality gaming in rehabilitation after stroke – user experiences and perceptions. *Disability and Rehabilitation* 44, 22 (Aug. 2021), 6759–6765. doi:10.1080/09638288.2021.1972351
- [15] Nadia Hocine, Abdelkader Gouaïch, and Stefano A. Cerri. 2014. *Dynamic Difficulty Adaptation in Serious Games for Motor Rehabilitation*. Springer International Publishing, 115–128. doi:10.1007/978-3-319-05972-3_13
- [16] Nadia Hocine, Abdelkader Gouaïch, Stefano A. Cerri, Denis Mottet, Jérôme Froger, and Isabelle Laffont. 2015. Adaptation in serious games for upper-limb rehabilitation: an approach to improve training outcomes. *User Modeling and User-Adapted Interaction* 25, 1 (Jan. 2015), 65–98. doi:10.1007/s11257-015-9154-6
- [17] Tetsunari Inamura, Satoshi Uenaka, Satoshi Shibuya, Yukari Ohki, Yutaka Oouchida, and Shin-ichi Izumi. 2016. Development of VR platform for cloud-based neurorehabilitation and its application to research on sense of agency and ownership. *Advanced Robotics* 31, 1–2 (Dec. 2016), 97–106. doi:10.1080/01691864.2016.1264885
- [18] Kenta Kamikokuryo, Takumi Haga, Gentiane Venture, and Vincent Hernandez. 2022. Adversarial Autoencoder and Multi-Armed Bandit for Dynamic Difficulty Adjustment in Immersive Virtual Reality for Rehabilitation: Application to Hand Movement. *Sensors* 22, 12 (June 2022), 4499. doi:10.3390/s22124499
- [19] Charles R. Kelley. 1969. What is Adaptive Training? *Human Factors: The Journal of the Human Factors and Ergonomics Society* 11, 6 (Dec. 1969), 547–556. doi:10.

- 1177/001872086901100602
- [20] Konstantina Kilteni, Raphaela Groten, and Mel Slater. 2012. The Sense of Embodiment in Virtual Reality. *Presence: Teleoperators and Virtual Environments* 21, 4 (Nov. 2012), 373–387. doi:10.1162/pres_a_00124
 - [21] Lasse F. Lui, Unnikrishnan Radhakrishnan, Francesco Chinello, and Konstantinos Koumaditis. 2025. The efficacy of adaptive training in immersive virtual reality for a fine motor skill task. *Virtual Reality* 29, 1 (Jan. 2025). doi:10.1007/s10055-024-01083-z
 - [22] Maria Grazia Maggio, Giuseppa Maresca, Rosaria De Luca, Maria Chiara Stagnitti, Bruno Porcari, Maria Cristina Ferrera, Franco Galletti, Carmela Casella, Alfredo Manuli, and Rocco Salvatore Calabrò. 2019. The Growing Use of Virtual Reality in Cognitive Rehabilitation: Fact, Fake or Vision? A Scoping Review. *Journal of the National Medical Association* 111, 4 (Aug. 2019), 457–463. doi:10.1016/j.jnma.2019.01.003
 - [23] Elisa Mantovani, Chiara Zucchella, Sara Bottiroli, Angela Federico, Rosalba Giugno, Giorgio Sandrini, Cristiano Chiamulera, and Stefano Tamburin. 2020. Telemedicine and Virtual Reality for Cognitive Rehabilitation: A Roadmap for the COVID-19 Pandemic. *Frontiers in Neurology* 11 (Sept. 2020). doi:10.3389/fneur.2020.00926
 - [24] Catherine A Mateer. 2005. Fundamentals of cognitive rehabilitation. *Effectiveness of rehabilitation for cognitive deficits* 21 (2005), 29.
 - [25] Fatemeh Mortazavi, Hadi Moradi, and Abdol-Hossein Vahabie. 2024. Dynamic difficulty adjustment approaches in video games: a systematic literature review. *Multimedia Tools and Applications* 83, 35 (March 2024), 83227–83274. doi:10.1007/s11042-024-18768-x
 - [26] Haruka Murakami, Vittorio Fiscale, Agata Marta Soccini, and Tetsunari Inamura. 2024. White Lies in Virtual Reality: Impact on Enjoyment and Fatigue. In *30th ACM Symposium on Virtual Reality Software and Technology (VRST '24)*. ACM, 1–2. doi:10.1145/3641825.3689692
 - [27] Domen Novak. 2018. *Promoting motivation during robot-assisted rehabilitation*. Elsevier, 149–158. doi:10.1016/b978-0-12-811995-2.00010-2
 - [28] Chiara Pagliari, Sonia Di Tella, Johanna Jonsdottir, Laura Mendozzi, Marco Rovaris, Roberto De Icco, Tommaso Milanese, Sara Federico, Michela Agostini, Michela Goffredo, Leonardo Pellicciari, Marco Franceschini, Vincenzo Cimino, Placido Bramanti, and Francesca Baglio. 2021. Effects of home-based virtual reality telerehabilitation system in people with multiple sclerosis: A randomized controlled trial. *Journal of Telemedicine and Telecare* 30, 2 (Dec. 2021), 344–355. doi:10.1177/1357633x211054839
 - [29] Masoud Rahimi, Hadi Moradi, Abdol-hossein Vahabie, and Hamed Kebriaei. 2023. Continuous Reinforcement Learning-based Dynamic Difficulty Adjustment in a Visual Working Memory Game. doi:10.48550/ARXIV.2308.12726
 - [30] William R Sherman and Alan B Craig. 2018. *Understanding virtual reality: Interface, application, and design*. Morgan Kaufmann.
 - [31] Agata Marta Soccini. 2020. The Induced Finger Movements Effect. In *SIGGRAPH Asia 2020 Posters (SA '20)*. ACM, 1–2. doi:10.1145/3415264.3425448
 - [32] Agata Marta Soccini and Federica Cena. 2021. The Ethics of Rehabilitation in Virtual Reality: the role of Self-Avatars and Deep Learning. In *2021 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR)*. IEEE, 324–328. doi:10.1109/aivr52153.2021.00068
 - [33] Agata Marta Soccini, Alessandro Clocchiatti, and Tetsunari Inamura. 2022. Effects of Frequent Changes in Extended Self-Avatar Movements on Adaptation Performance. *Journal of Robotics and Mechatronics* 34, 4 (Aug. 2022), 756–766. doi:10.20965/jrm.2022.p0756
 - [34] Agata Marta Soccini, Marco Grangetto, Tetsunari Inamura, and Sotaro Shimada. 2019. Virtual Hand Illusion: The Alien Finger Motion Experiment. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 1165–1166. doi:10.1109/vr.2019.8798193
 - [35] Paweł Strojny and Natalia Dużmańska-Misiarczyk. 2023. Measuring the effectiveness of virtual training: A systematic review. *Computers & Education: X Reality* 2 (2023), 100006. doi:10.1016/j.cexr.2022.100006
 - [36] Joanna Szczepańska-Gieracha and Justyna Mazurek. 2020. The Role of Self-Efficacy in the Recovery Process of Stroke Survivors. *Psychology Research and Behavior Management* Volume 13 (Nov. 2020), 897–906. doi:10.2147/prbm.s273009
 - [37] Liu Wang, Mengjie Huang, Rui Yang, Hai-Ning Liang, Ji Han, and Ying Sun. 2023. Survey of Movement Reproduction in Immersive Virtual Rehabilitation. *IEEE Transactions on Visualization and Computer Graphics* 29, 4 (April 2023), 2184–2202. doi:10.1109/tvcg.2022.3142198
 - [38] Maryam Zahabi and Ashiq Mohammed Abdul Razak. 2020. Adaptive virtual reality-based training: a systematic literature review and framework. *Virtual Reality* 24, 4 (March 2020), 725–752. doi:10.1007/s10055-020-00434-w
 - [39] Mohammad Zohaib. 2018. Dynamic Difficulty Adjustment (DDA) in Computer Games: A Review. *Advances in Human-Computer Interaction* 2018 (Nov. 2018), 1–12. doi:10.1155/2018/5681652